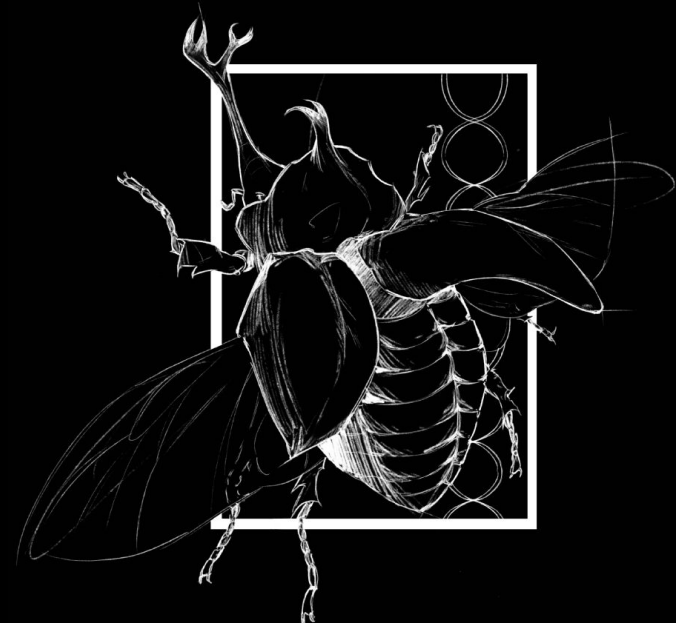




Elytral Black Coloration and Habitat Preference in Ladybugs: Habitat Thermoregulation Trade-off in Coccinellidae

MacKenzie Wilkerson and Heath Blackmon
Department of Biology, Texas A&M University, College Station, TX, USA



Background

Color variation in ladybugs (Coccinellidae) is thought to be influenced by ecological and environmental pressures, particularly temperature and habitat conditions. In many insects, darker coloration is associated with thermoregulation advantages in cooler environments, suggesting a potential trade-off between coloration and habitat preference. **This study examines variation in elytral black coloration (percent dorsal black surface) across species of Coccinellidae to test whether coloration patterns are associated with geographic gradients (latitude and longitude) and habitat type.** Additionally, these traits are considered within a phylogenetic context to explore how evolutionary relationships may influence observed patterns. The goal of this project is to evaluate whether elytral coloration is linked to environmental conditions, consistent with a potential thermoregulation–habitat trade-off.

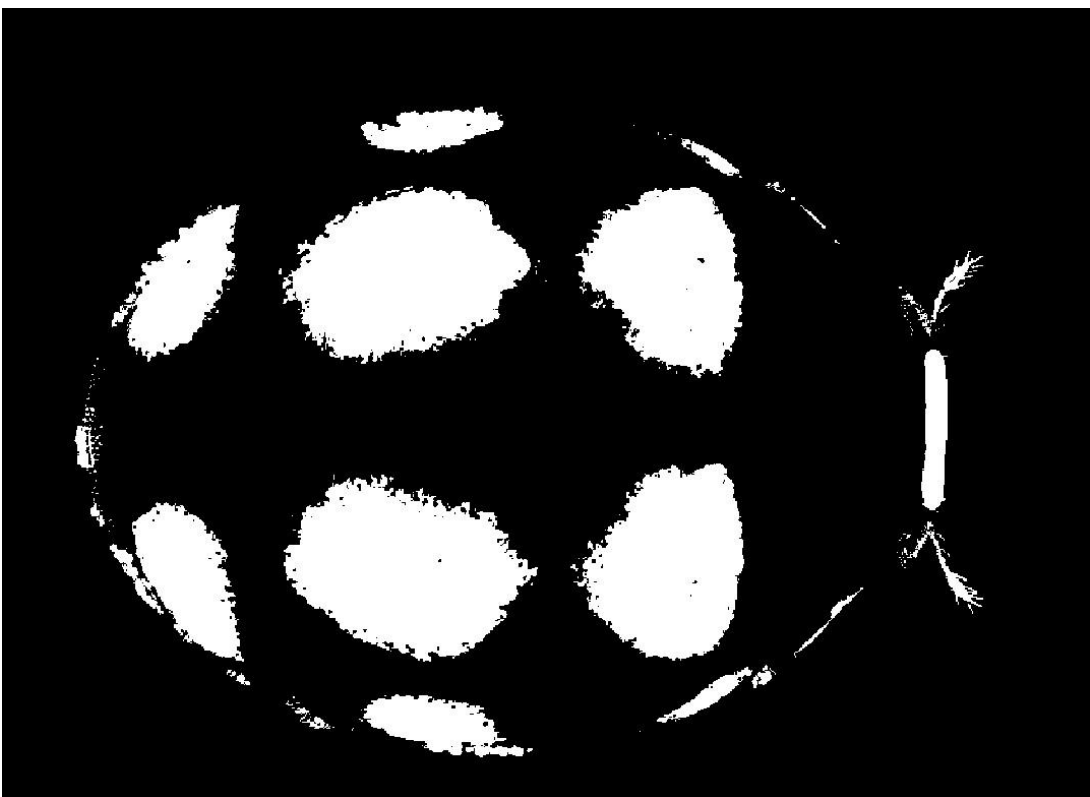
Study Design & AI

This study compiled a dataset of ladybug (Coccinellidae) species including genus, species, geographic location (latitude and longitude), habitat type, and elytral black coloration (percent dorsal black surface). Coloration was quantified using ImageJ and Fiji by applying thresholding to images to estimate percent black area.

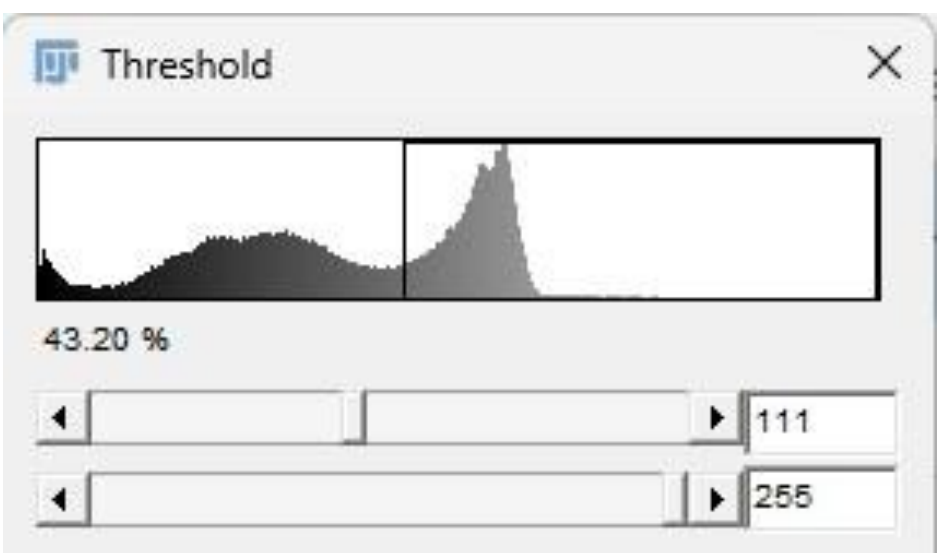
The collected data was cleaned and analyzed using an application called R. Linear models were used to test relationships between percent black coloration and geographic variables (latitude and longitude) and as well as habitat type. Scatterplots and regression lines were generated to visualize these relationships. A phylogenetic tree was also used to visualize variation in coloration across evolutionary lineages between different genera in Coccinellidae

AI tools were used to assist with debugging R code, organizing data analysis workflows, and improving clarity of written explanations. All analyses, interpretations, and final decisions were performed by the author.

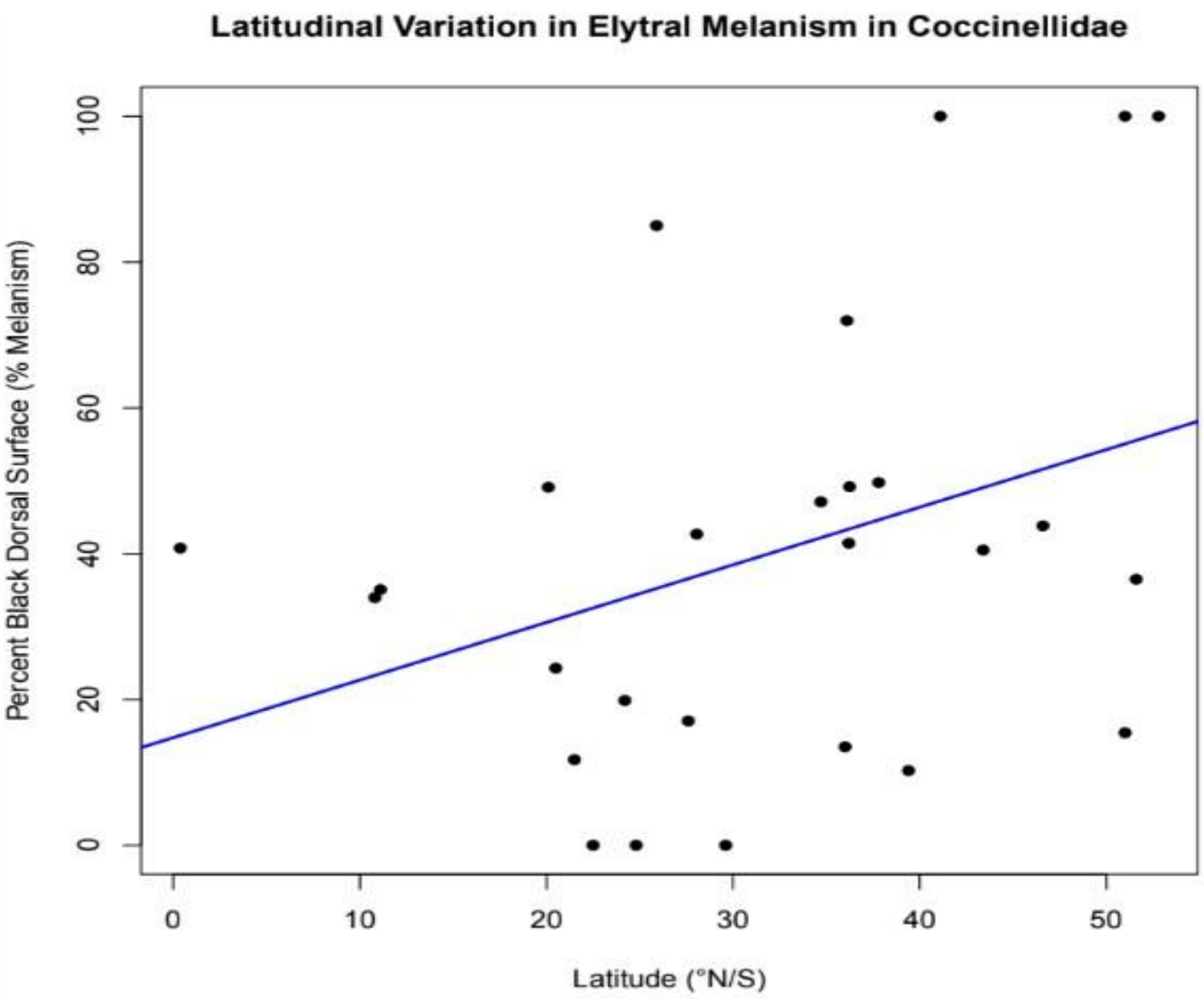
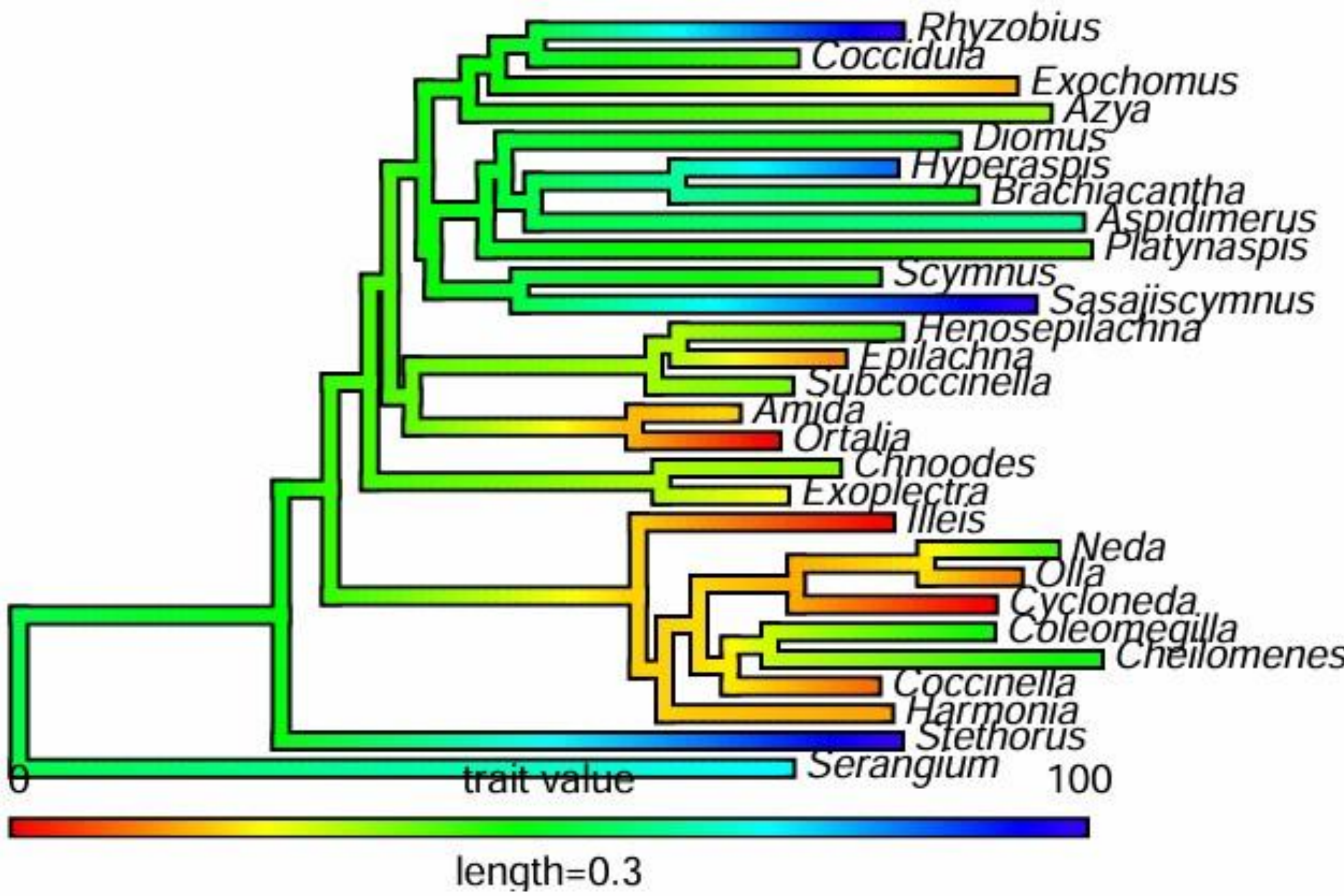
What does the data look like?



	Area	%Area
Brachiacantha felina	210409	43.427



Results



Percent black elytral coloration showed a weak but significant positive relationship with latitude, indicating that individuals at higher latitudes tended to have darker coloration. In contrast, longitude was not a significant predictor of coloration, and habitat type did not show a significant effect, so these models are not shown. Overall, the models explained a relatively small portion of the variation in coloration, suggesting substantial variability within the dataset.

Discussion

This study investigated whether elytral black coloration varies with geographic gradients and habitat type in ladybugs (Coccinellidae), and whether these patterns are consistent with a potential thermoregulation and habitat trade-off.

Overall, the results show a weak but significant positive relationship between latitude and percent black coloration, suggesting that individuals from higher latitudes tend to have darker elytra. This pattern is consistent with the idea that increased dark pigmentation may provide thermoregulatory advantages in cooler environments. However, latitude only explains a modest portion of variation in coloration, indicating that additional factors likely contribute to observed differences.

In contrast, longitude was not a significant predictor of coloration, and habitat type did not show a consistent or significant effect in any of the models. This suggests that broad habitat categories (e.g., forest, agriculture, urban) may not capture the ecological drivers influencing elytral coloration in this dataset.

The phylogenetic analysis provides additional context by showing how melanism is distributed across evolutionary lineages, suggesting that shared ancestry may also contribute to variation in coloration. However, further comparative analysis would be needed to separate phylogenetic effects from environmental influences.

Overall, these findings suggest that while there is some evidence for a latitudinal gradient in elytral coloration, the relationship is weak and likely influenced by multiple ecological and evolutionary factors rather than a single environmental driver.

Conclusions & Acknowledgements

This study examined whether elytral black coloration is associated with habitat and geographic variation in ladybugs (Coccinellidae). Results indicate a weak but significant positive relationship between latitude and percent black coloration, suggesting increased melanization at higher latitudes. In contrast, longitude and habitat type were not significant predictors of coloration.

Overall, these findings provide limited support for a thermoregulation-related pattern, but suggest that coloration is influenced by multiple factors beyond broad habitat categories or simple geographic gradients. Additional variables, such as microhabitat conditions or species-specific traits, may play an important role in shaping coloration patterns I would like to thank Dr. Blackmon for guidance and support throughout this project, as well as the BIOL 491 course for providing the opportunity to conduct independent research. I also acknowledge the use of publicly available image and biodiversity databases (e.g., museum collections and online repositories) that made this analysis possible.